JOEL HILMERSSON

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About

Versatile ***computational designer*** and ***software developer*** with a background in architecture and engineering, currently working within the field of software for design and manufacturing. My main interest lies in computational tools for ***geometry modeling***, ***fabrication*** and ***design space exploration***, and how we can leverage technology to turn innovative design concepts into physical reality.

Professional Experience

Aibuild | London, UK
Early Series-A hire; core member of a three-person computational geometry team responsible for extending and maintaining a custom slicing engine for multi-axis robotic additive manufacturing. (Java/C++)

Senior Geometry Engineer | May 2024 – Present
• Own end-to-end development of geometry driven features (e.g., ***metal part repair from 3D scans***, ***parametric design tools***) in close collaboration with R&D and platform teams.
• Developed a scalar-field based ***density model*** for the ***as-deposited geometry***, to analyse extrusion and detect porosity, overfill and other defects, enabling ***closed-loop feedback*** for toolpath generation and optimisation.
• Integrated external libraries into the software via JNI, incl. ***CGAL*** and ***PCL***, enabling ***advanced mesh*** and ***point-cloud processing*** capabilities.

Computational Geometry Engineer | June 2023 – April 2024
• Created algorithms for complex, ***geometry-aware slicing*** using nonplanar and adaptive strategies, enabling applications for clients in construction, automotive, and aerospace industries.
• Implemented a diverse range of geometric functionality (covering ***graphs & topology***, ***UV parameterization***, ***scalar and vector fields*** and ***implicit geometry***) from papers and technical sources into core components of the engine.

Key Skills: Java, C++, Computational Geometry, Additive Manufacturing, R&D

Generative Engineering | London, UK
Start-up company developing a platform for generative design in engineering. I joined the company in pre-seed.

Computational Engineer | Jan 2023 – April 2023
• Explored the application of generative models for design applications.
• PoC integration of cloud-based FE simulations via SimScale's API. (Python)

Key Skills: Python, Generative Design, Multi-Objective Optimisation

AKT II | Applied Research | London, UK
Worked as part of the software development team, which is 5 person a sub-group of the wider interdisciplinary applied research team.

Computational Designer | Feb 2020 – Jan 2023
• Lead the development of ***Reakt***, an ***in-house interoperability toolkit*** for parametric workflows, ***integrating design (Rhino/Gh) and analysis software*** through a bespoke object model (C#).
• Supported projects with ***finite element analysis (FEA)*** and ***complex parametric modeling***, collaborating with leading architects such as Foster + Partners and Zaha Hadid Architects.
• Contributed to the ***Horizon 2020 PrismArch*** project as a domain expert.
• Conducted internal R&D on meshing for FEA, workflow automation, and computational modeling, providing custom tools for project teams. (C#, C++)

Key Skills: Rhino, Grasshopper, NURBS, C#, Python, FEA Software (+APIs)

Internships & Part time work

• Sunneroe Architects | Gothenburg, Sweden | 2019
Part-time external consultant. Developed a generative grasshopper toolkit for automated apartment layout design. Using Rhino, Grasshopper, Python.

• Bollinger + Grohmann | Oslo, Norway | 2018
FEA Analysis & Multi-Objective Optimization.
Using Rhino, Grasshopper, C#, Karamba, RFem.

• Knippers Helbig | Stuttgart, Germany | 2015 - 2016
Computational design for complex geometry and FEA for projects globally.
Using Rhino, Grasshopper, C#, Sofistik.

Education

Chalmers University of Technology | Gothenburg, Sweden

M.Arch | Architecture & Urban Design | Sept 2017 - Aug 2019
• Master in architecture & urban design with a focus on digital design and robotic fabrication.

M.Sc | Structural Engineering & Building Technology | Sept 2016 – Aug 2019
• Master in engineering with a focus on structural mechanics and FEA.
• Thesis: Isogeometric analysis & form finding
Wrote a design tool for form active thin shells using NURBS based finite element analysis (Isogeometric Analysis). (Grasshopper, MATLAB, C#)

B.Sc | Architecture & Engineering | Aug 2012 – June 2015
• Interdisciplinary bachelor's degree, combining design studies with engineering and maths.

Delft University of Technology | Delft, Netherlands

Erasmus | Computational Mechanics | Sept 2016 – Aug 2017
• 1 Year ERASMUS scholarship as part of engineering master.
Studies in engineering specializing in computational mechanics.

Academic & Research Projects

PrismArch | Horizon2020 | 2020 – 2023
An interdisciplinary research project bridging academia and industry to explore VR-based design workflows. Collaborators included Zaha Hadid Architects, ETH Zurich, and the University of Malta. Funded by the EU's Horizon 2020 program.

• Applied research on evolutionary algorithms in collaboration with AI researchers at the University of Malta, resulting in a shape optimization framework for structural design (C#).

• Authored key sections of Horizon 2020 deliverables, on the topics of cross-disciplinary AEC ontologies and algorithmic design principles.

Conference Papers

• IASS 2023 | *Design Space Exploration of Shell Structures Using Quality Diversity Algorithms*
K. Sfikas, A, Liapis, J. Hilmersson, J. Dudley, E. Tibuzzi, G. Yannakakis

• IASS 2021 | *The Geldeford Riband*
D. Godfrey, J. Dudley, J. Hilmersson

• IASS 2019 | Isogeometric Analysis and Form Finding of Thin Elastic Shells
J. Hilmersson, J. Olsson, M. Ander, F. Larsson

Tutoring | Chalmers University of Technology | 2015 – 2019
Courses: Space & Geometry • Structural Mechanics • Solid Mechanics • Digital & Parametric Design

Technical Skills

Programming languages: C#, java, python, C++, rust
Libraries/Tools: eigen, CGAL, OpenVDB (familiar), JNI, cmake

Simulation & FEA: Sofistik, SAP2000, ETABS, RFem (+APIs)
3D modeling: Rhino, Grasshopper, Blender

Domain Expertise:
computational geometry (*nurbs*, *implicit*), parametric modeling, finite element analysis, design space exploration & optimisation, additive manufacturing